

Background Exercises

Solution

AE 513: Multivariable Control — Autumn 2025

Linear Algebra

Eigenvalue Basics

Q1. Let $A = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$. What are the eigenvalues of A ?

- (a) $\{4, 2\}$
- (b) $\{6, -1\}$
- (c) $\{1, 1\}$
- (d) $\{4, 1\}$

Solution. Answer: (d) $\{4, 1\}$.

Why: For an upper/lower triangular matrix, the eigenvalues are exactly the diagonal entries. Formally, $\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 4 & -2 \\ 0 & \lambda - 1 \end{bmatrix} = (\lambda - 4)(\lambda - 1)$, whose roots are $\lambda = 4, 1$.

Q2. Let $B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Which vector is an eigenvector with eigenvalue 4?

- (a) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- (b) $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
- (c) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Solution. Answer: (d) $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ with $\lambda = 4$.

Definition: $v \neq 0$ is an eigenvector of B with eigenvalue λ if $Bv = \lambda v$. Check: $B \begin{bmatrix} 1 \\ 1 \end{bmatrix} =$

$$\begin{bmatrix} 3 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Shapes & Transpose

Q3. Let $A \in \mathbb{R}^{2 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$, $x \in \mathbb{R}^2$. Which statement is correct?

- (a) $AB \in \mathbb{R}^{3 \times 3}$; $BA \in \mathbb{R}^{2 \times 2}$; Ax defined; Bx undefined.
- (b) $AB \in \mathbb{R}^{2 \times 2}$; $BA \in \mathbb{R}^{3 \times 3}$; Ax undefined; $Bx \in \mathbb{R}^3$.
- (c) $AB \in \mathbb{R}^{2 \times 3}$; $BA \in \mathbb{R}^{3 \times 2}$; both Ax, Bx defined.
- (d) Only Ax is defined.

Solution. Answer: (b).

Shapes: $A \in \mathbb{R}^{2 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$, $x \in \mathbb{R}^2$. Then $AB \in \mathbb{R}^{2 \times 2}$ and $BA \in \mathbb{R}^{3 \times 3}$. The product Ax is undefined since inner dimensions 3 (of A) and 2 (of x) do not match; $Bx \in \mathbb{R}^3$ is defined because $B(3 \times 2)$ times $x(2 \times 1)$ yields (3×1) .

Q4. Which identity holds for any conformable A, B ?

- (a) $(AB)^\top = A^\top B^\top$
- (b) $(AB)^\top = B^\top A^\top$
- (c) $(AB)^\top = AB$
- (d) None of the above

Solution. Answer: (b).

Identity: $(AB)^\top = B^\top A^\top$.

Reason: The transpose reverses the order of multiplication while transposing each factor.

Definiteness & Symmetry

Q5. Determine the definiteness of the matrix $D = \text{diag}(2, 5)$.

- (a) Positive-Semi-Definite
- (b) Positive-Definite
- (c) Negative-Definite
- (d) Indefinite

Solution. Answer: (b) Positive-Definite (PD).

Definition: A symmetric matrix D is PD iff $x^\top D x > 0$ for all $x \neq 0$.

For diagonal $D = \text{diag}(d_1, d_2)$, PD $\Leftrightarrow d_1 > 0$ and $d_2 > 0$. Here $2, 5 > 0$.

Q6. For any real M , which statements are always true?

- (a) $M + M^\top$ is symmetric
- (b) $M - M^\top$ is symmetric
- (c) $(M + M^\top)^\top = M + M^\top$
- (d) Both (a) and (c)

Solution. Answer: (d) Both (a) and (c).

Facts: $M + M^\top$ is symmetric since $(M + M^\top)^\top = M + M^\top$. $M - M^\top$ is skew-symmetric since $(M - M^\top)^\top = -(M - M^\top)$.

Extra

Q7. Let T be invertible and $C = TAT^{-1}$. Which are always equal for A and C ?

- (a) trace and determinant
- (b) trace only
- (c) determinant only
- (d) all entries

Solution. Answer: (a) trace and determinant.

Key point: If $C = TAT^{-1}$ with T invertible, then A and C are similar.

Similarity preserves eigenvalues, thus also $\text{tr}(\cdot)$ (sum of eigenvalues) and $\det(\cdot)$ (product of eigenvalues), but not individual entries.

Q8. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. What is $\text{rank}(A)$?

- (a) 0
- (b) 1
- (c) 2
- (d) 3

Solution. Answer: (b) $\text{rank} = 1$.

Reason: Row 2 = $2 \times$ Row 1, so only one linearly independent row/column.

Definition: $\text{rank}(A)$ = dimension of the column space = number of independent columns (or rows).

Q9. Which matrix is orthogonal?

- (a) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Solution. Answer: (a).

Definition: Q is orthogonal if $Q^\top Q = I$ (columns are orthonormal).

For $Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $Q^\top Q = I$. Others fail $Q^\top Q = I$ (e.g., scaling or shear).

Q10. What is D^{-1} for $D = \text{diag}(2, 5)$?

- (a) $\text{diag}(2, 5)$
- (b) $\text{diag}(5, 2)$
- (c) $\text{diag}\left(\frac{1}{5}, \frac{1}{2}\right)$
- (d) $\text{diag}\left(\frac{1}{2}, \frac{1}{5}\right)$

Solution. Answer: (d) $\text{diag}\left(\frac{1}{2}, \frac{1}{5}\right)$.

Rule: $\left(\text{diag}(d_1, d_2)\right)^{-1} = \text{diag}(1/d_1, 1/d_2)$ in the same order, hence $1/2, 1/5$.

Multivariable Calculus

Gradients

Q1. For $f(x, y) = x^2y$, compute $\nabla f(x, y)$.

- (a) $(x^2, 2xy)$
- (b) $(2xy, x^2)$
- (c) $(2x, y)$
- (d) $(2x, 2y)$

Solution. Answer: (b) $(2xy, x^2)$.

Definition: $\nabla f = (\partial f / \partial x, \partial f / \partial y)$.

For $f(x, y) = x^2y$: $f_x = 2xy$ (treat y constant), $f_y = x^2$ (treat x constant).

Q2. For $f(x, y) = e^xy^2$, compute $\nabla f(x, y)$.

- (a) $(y^2e^x, 2ye^x)$
- (b) $(e^x, 2y)$
- (c) $(y^2, 2y)$
- (d) $(y^2, 2ye^x)$

Solution. Answer: (a) $(y^2e^x, 2ye^x)$.

Because $f(x, y) = e^xy^2$: $f_x = y^2e^x$ (product where y^2 is constant wrt x), $f_y = 2ye^x$.

Jacobians

Q3. For $F(x, y) = (x^2, xy)$, find the Jacobian $J_F(x, y)$.

- (a) $\begin{bmatrix} 2x & 0 \\ y & x \end{bmatrix}$
- (b) $\begin{bmatrix} 2x & y \\ 0 & x \end{bmatrix}$
- (c) $\begin{bmatrix} x & 0 \\ y & 2x \end{bmatrix}$
- (d) $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

Solution. Answer: (a).

Definition: $J_F = \begin{bmatrix} \partial F_1 / \partial x & \partial F_1 / \partial y \\ \partial F_2 / \partial x & \partial F_2 / \partial y \end{bmatrix}$.

With $F_1 = x^2 \Rightarrow (2x, 0)$ and $F_2 = xy \Rightarrow (y, x)$, hence $J_F = \begin{bmatrix} 2x & 0 \\ y & x \end{bmatrix}$.

Q4. For $G(x, y, z) = (x + y, x + z)$, compute $J_G(x, y, z)$.

- (a) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$
- (c) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Solution. Answer: (a).

For $G(x, y, z) = (x + y, x + z)$, the coefficient matrix of the linear map is $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, which is J_G .

Second Derivatives & Hessians

Q5. For $f(x, y) = 3x^2 + 2xy + y^2$, compute the Hessian $Hf(x, y)$.

- (a) $\begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix}$
- (b) $\begin{bmatrix} 6 & 1 \\ 2 & 2 \end{bmatrix}$
- (c) $\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$
- (d) $\begin{bmatrix} 6 & 2 \\ 1 & 2 \end{bmatrix}$

Solution. Answer: (a).

Definition: $Hf = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$.

For $f = 3x^2 + 2xy + y^2$: $f_{xx} = 6$, $f_{xy} = f_{yx} = 2$, $f_{yy} = 2 \Rightarrow Hf = \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix}$.

Q6. For $f(x, y) = x^3 + y^2$, compute the Hessian $Hf(x, y)$.

- (a) $\begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}$
- (b) $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$
- (c) $\begin{bmatrix} 3x^2 & 0 \\ 0 & 2y \end{bmatrix}$
- (d) $\begin{bmatrix} 6x & 3x^2 \\ 0 & 2 \end{bmatrix}$

Solution. Answer: (a).

For $f = x^3 + y^2$: $f_{xx} = 6x$, $f_{xy} = 0$, $f_{yy} = 2 \Rightarrow Hf = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}$.

Extra Concepts

Q7. For $f(x, y) = x^2 + y^2$, find the critical point(s) and classify.

- (a) No critical points; function increases everywhere
- (b) Single critical point at $(0, 0)$; strict local minimum
- (c) Single critical point at $(0, 0)$; strict local maximum
- (d) Single critical point at $(0, 0)$; saddle point

Solution. Answer: (b) strict local minimum at $(0, 0)$.

Steps: $\nabla f = (2x, 2y) = 0 \Rightarrow (x, y) = (0, 0)$.

Second-derivative test: $Hf = 2I \succ 0$ (positive definite) implies a strict local minimum.

Q8. For $f(x, y) = xe^y$, compute $\frac{\partial f}{\partial x}(1, 0)$.

- (a) 0
- (b) 1
- (c) e
- (d) e^1

Solution. Answer: (b) 1.

Reason: $f(x, y) = xe^y \Rightarrow f_x = e^y$. Evaluate at $(1, 0)$: $e^0 = 1$.

Q9. Let $x = r \cos \theta$, $y = r \sin \theta$. For $f(x, y) = x^2 + y^2$, compute $\frac{\partial}{\partial r} f(x(r, \theta), y(r, \theta))$.

- (a) 0
- (b) r
- (c) 2
- (d) $2r$

Solution. Answer: (d) $2r$.

Chain rule in polar: With $x = r \cos \theta$, $y = r \sin \theta$, we have $f = x^2 + y^2 = r^2$. Therefore $\partial f / \partial r = \partial(r^2) / \partial r = 2r$.

Q10. For the map $(u, v) \mapsto (x, y) = (2u, 3v)$, compute $\det(\partial(x, y) / \partial(u, v))$.

- (a) 0
- (b) 1
- (c) 6
- (d) -6

Solution. Answer: (c) 6.

Jacobian determinant: For $(u, v) \mapsto (x, y) = (2u, 3v)$, $J = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and $\det J = 2 \cdot 3 = 6$.
Geometric meaning: area scales by 6.

Optimization

Solve for the Minimizer

Q1. Minimize $f(x) = x^2 + 2x + 1$ over $x \in \mathbb{R}$. What is the minimizer x^* ?

- (a) -1
- (b) 0
- (c) 1
- (d) No minimizer

Solution. Answer: (a) $x^* = -1$.

Complete the square: $f(x) = x^2 + 2x + 1 = (x + 1)^2$. The minimum value 0 occurs at $x = -1$. Convexity: $f''(x) = 2 > 0$ implies a unique global minimizer.

Q2. Minimize $f(x, y) = (x - 1)^2 + (y + 2)^2$ over $\mathbb{R}^2$. What is (x^*, y^*) ?

- (a) $(1, -2)$
- (b) $(-1, 2)$
- (c) $(1, 2)$
- (d) $(0, -2)$

Solution. Answer: (a) $(x^*, y^*) = (1, -2)$.

Reason: $f(x, y) = (x - 1)^2 + (y + 2)^2$ is a sum of squares centered at $(1, -2)$. Gradient is zero there and Hessian $2I \succ 0$, so it is the unique global minimizer.

Q3. For $f(x, y) = x^2 + y^2 + x$, solve $\nabla f = 0$ and give the minimizer.

- (a) $(\frac{1}{2}, 0)$
- (b) $(-\frac{1}{2}, 0)$
- (c) $(0, 0)$
- (d) $(0, -\frac{1}{2})$

Solution. Answer: (b) $(x^*, y^*) = (-\frac{1}{2}, 0)$.

Derivation: $\nabla f = (2x + 1, 2y) = 0 \Rightarrow x = -\frac{1}{2}, y = 0$. Hessian $H = 2I \succeq 0$ (convex quadratic), hence global minimizer.

Convexity & Simple Constraints

Q4. For $f(x) = ax^2 + bx + c$ with $a > 0$, which statement is true?

- (a) f is concave on $\mathbb{R}$ and has no minimizer
- (b) f is convex on $\mathbb{R}$ and has a unique global minimizer $x^* = -\frac{a}{2b}$
- (c) f is convex on $\mathbb{R}$ and has a unique global minimizer $x^* = -\frac{b}{2a}$
- (d) f is neither convex nor concave

Solution. Answer: (c).

For $f(x) = ax^2 + bx + c$ with $a > 0$, f is convex on $\mathbb{R}$. Setting $f'(x) = 2ax + b = 0$ yields $x^* = -\frac{b}{2a}$ (unique global minimizer).

Q5. Minimize $(x - 3)^2$ subject to $x \geq 5$. What is the minimizer x^* ?

- (a) 3
- (b) 4
- (c) 5
- (d) No minimizer

Solution. Answer: (c) $x^* = 5$.

Unconstrained optimum $x = 3$ violates $x \geq 5$. Projection principle: For a convex function over a half-line, the constrained minimizer is the closest feasible point to the unconstrained minimizer, i.e., $x = 5$.

Q6. Which function is convex on $\mathbb{R}$?

- (a) $-x^2$
- (b) e^x
- (c) x^3
- (d) $\sin x$

Solution. Answer: (b) e^x .

Reason: $f''(x) = e^x > 0$ for all x , so f is strictly convex on $\mathbb{R}$. Others are not globally convex: $-x^2$ is concave; x^3 changes curvature; $\sin x$ oscillates.

Basic Python

Core Python

Q1. What does this print?

```
def add(a, b):  
    print(a + b)  
  
x = add(2, 3)  
print(x)
```

- (a) Prints 5 only
- (b) Prints None only
- (c) Prints 5 then None
- (d) Raises an error

Solution. Answer: (c).

Explanation: `add(2,3)` prints 5 but returns `None` (no `return` statement). Thus `x` is `None`, and `print(x)` prints `None`.

Q2. Let `nums = [10, 20, 30, 40, 50]`. What is `nums[4:0:-2]`?

- (a) `[50, 30]`
- (b) `[50, 30, 10]`
- (c) `[40, 20]`
- (d) Error

Solution. Answer: (a) `[50, 30]`.

Rule: Slice `[start:stop:step]` excludes `stop`. From index 4 down by `-2`: indices 4, 2 give values `[50, 30]`.

Q3. What does this print?

```
def append_item(x, lst=[]):  
    lst.append(x)  
    return lst  
  
print(append_item(1))  
print(append_item(2))
```

- (a) `[1]` then `[2]`
- (b) `[1]` then `[1, 2]`
- (c) Two separate lists: `[1]` then `[2]`
- (d) Error

Solution. Answer: (b).

Gotcha: Default mutable argument `lst=[]` is created once at function definition and shared across calls. So the second call appends to the same list, producing `[1]` then `[1,2]`.

Q4. What does this print?

```
m = [[0]*3]*2
m[0][1] = 9
print(m)
```

- (a) `[[0, 9, 0], [0, 0, 0]]`
- (b) `[[0, 9, 0], [0, 9, 0]]`
- (c) `[[0, 0, 0], [0, 0, 0]]`
- (d) Error

Solution. Answer: (b).

Reason: `[[0]*3]*2` duplicates the same inner list reference; modifying one row affects both, so the printed matrix is `[[0,9,0], [0,9,0]]`.

Q5. What does this print?

```
a = [1, 2, 3]
b = a
a.append(4)
print(b)
```

- (a) `[1, 2, 3]`
- (b) `[1, 2, 3, 4]`
- (c) None
- (d) Error

Solution. Answer: (b).

Reason: `b=a` makes `b` reference the same list as `a`. Appending via `a` mutates that object, so printing `b` shows `[1,2,3,4]`.

NumPy Essentials

Q6. (Column vector) You have `v = [1, 2, 3]`. Create a NumPy column vector with shape `(3, 1)`. Which snippet does that?

- (a) `np.array(v)`
- (b) `np.reshape(v, (1, 3))`
- (c) `np.array(v).T`
- (d) `np.array(v).reshape(-1, 1)`

Solution. Answer: (d) `np.array(v).reshape(-1, 1)` gives shape `(3,1)`.

Notes: `np.array(v)` is 1D $(3,)$; its transpose is still $(3,)$. Reshaping produces a true column vector.

Q7. (Matmul shape) If `A.shape == (2, 3)` and `x.shape == (3,)`, what is the shape of `A @ x`?

- (a) $(2,)$
- (b) $(2, 1)$
- (c) $(1, 2)$
- (d) Error

Solution. Answer: (a) shape $(2,)$.

Rule: In NumPy, $(2 \times 3)@(3,) \rightarrow (2,)$ (a 1D result with the leading dimension of the matrix).